

**FAST NUCES**

Software Engineering

**PRISM**

*Presence Real-time Identification & Security Model*

**Deliverable #2 - Iteration 1 / Sprint 1**

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# 1. Project Introduction

**Project Title:** PRISM - Presence Real-time Identification & Security Model

**Type:** Web Application (React, Node.js, MongoDB)

## 1.1 Problem Statement

With the shift toward online education, verifying genuine student presence has become difficult. Traditional methods like passwords or login timestamps are easy to abuse through proxy attendance. Many existing systems use facial recognition only, which does not work for niqab-wearing students, or rely on GPS tracking, which raises privacy concerns. There is a need for a secure and inclusive attendance verification system.

## 1.2 Proposed Solution

PRISM is a web-based attendance verification system that uses facial recognition and fingerprint authentication. It includes fraud prevention through liveness checks and a confidence score. Students can choose their preferred verification method, making it more inclusive. The system runs entirely in the browser and does not use location tracking.

## 1.3 Technology Stack

- Frontend: React.js with Tailwind CSS
- Backend: Node.js with Express.js
- Database: MongoDB
- AI Service: Python (facial recognition and fingerprint verification) - planned for Sprint 2
- Deployment: Vercel (frontend) + Render (backend)

## 2. Sprint Backlog - Sprint 1

### 2.1 Module: UI Foundation & Frontend Architecture

This sprint focuses on building the complete frontend foundation of PRISM - all screens, navigation flows, and role-based dashboard layouts. This includes the user registration and login UI, role-specific dashboards for students, instructors, and admins, and the biometric enrollment interface. The backend project structure and MongoDB connection have also been established. Business logic and real data integration are planned for Sprint 2.

### 2.2 Sprint Backlog Table

Sprint Duration: 2 Weeks | Total Estimated Hours: 30 | Sprint Goal: Complete frontend with role-based dashboards, navigation, and backend project scaffold.

| Story ID | Priority | User Story Title                      | Est. Hrs | Status    | Sprint   |
|----------|----------|---------------------------------------|----------|-----------|----------|
| US-01    | High     | User Registration & Authentication UI | 8        | Done (UI) | Sprint 1 |
| US-02    | High     | Biometric Enrollment Interface (UI)   | 10       | Done (UI) | Sprint 1 |
| US-03    | High     | Instructor Dashboard & Session UI     | 7        | Done (UI) | Sprint 1 |
| US-04    | Medium   | Theme Switching (Light/Dark)          | 5        | Leftover  | Sprint 2 |

*Note: Sprint 1 focused on establishing the full frontend architecture. All UI screens and navigation flows are complete and functional. Backend API logic and real data integration are deferred to Sprint 2, following a frontend-first development strategy to validate UX before writing business logic.*

## 3. User Stories

### 3.1 US-01 - User Registration & Authentication UI

**Priority:** High

**Estimated Hours:** 8

**Status:** Done (UI Complete - Backend Logic in Sprint 2)

**User Story:**

As an administrator, I want to register and log in with role-based access so that the system can identify me and take me to the right dashboard.

**Description:**

This covers the complete UI for registration and login flows across all user roles - students, instructors, and admins. The screens include form validation feedback, role selection, and navigation to role-specific dashboards after login. In Sprint 1, navigation is handled via frontend routing with hardcoded role state. Real JWT-based authentication and backend validation will be completed in Sprint 2.

**Sub User Stories:**

- As a student, I want to register with my university email so I can access the system.
- As an instructor, I want to log in with my credentials so I am taken to the instructor dashboard.
- As an admin, I want to manage user accounts so I can control who has access.
- As a user, I want to see clear error messages when my login fails.

**Acceptance Criteria (Sprint 1 - UI Scope):**

- Registration form renders with full name, email, password, and role fields.
- Password validation UI shows requirements (8+ characters, uppercase, digit).
- Users are navigated to their role-specific dashboard after selecting a role.
- Login page renders with email and password fields and a 'Forgot?' link.
- Error state UI is shown for invalid login attempts.
- All screens are responsive and work on desktop and mobile browsers.

### 3.2 US-02 - Biometric Enrollment Interface (UI)

**Priority:** High

**Estimated Hours:** 10

**Status:** Done (UI Complete — Biometric Logic in Sprint 2)

**User Story:**

As a student, I want to enroll my biometric data (face and/or fingerprint) so that I can verify my attendance without using shared credentials.

**Description:**

This covers the UI for the biometric enrollment screen. The interface guides the student step by step through choosing an enrollment method (face, fingerprint, or both), and shows the appropriate capture interface for each. In Sprint 1, the webcam feed placeholder, alignment guides, and enrollment method selection are all built as UI components. Actual biometric capture, template extraction, and backend storage will be implemented in Sprint 2 when the AI service is integrated.

**Sub User Stories:**

- As a student, I want to choose between face, fingerprint, or both enrollment modes.
- As a student, I want step-by-step guidance during the enrollment process.
- As a student, I want confirmation that my biometric data was saved successfully.
- As a student, I want to re-enroll if my initial attempt fails quality checks.

**Acceptance Criteria (Sprint 1 - UI Scope):**

- Enrollment screen is accessible from the student dashboard.
- UI presents three enrollment options: face-only, fingerprint-only, or both.
- Face enrollment screen shows a webcam placeholder with alignment guide overlay.
- Step indicator shows current progress through the enrollment steps.
- A success confirmation UI state is shown upon simulated completion.
- A re-enroll option is available in the UI for failed attempts.

### 3.3 US-03 - Instructor Dashboard & Session UI

**Priority:** High

**Estimated Hours:** 7

**Status:** Done (UI Complete - Session Logic in Sprint 2)

**User Story:**

As an instructor, I want to create and manage attendance sessions so that students can mark attendance within a set time window.

**Description:**

This covers the instructor dashboard UI. The screen shows a course list, a session management table with dates and time slots, a flagged cases panel showing students with low confidence scores, and an Add New Session button. In Sprint 1, the dashboard renders with representative hardcoded data to validate the layout and information architecture. Real-time session creation and live attendance tracking will be implemented in Sprint 2.

**Sub User Stories:**

- As an instructor, I want to see all my assigned courses from my dashboard.
- As an instructor, I want to create a new attendance session with a start and end time.
- As an instructor, I want to see live progress of who has marked attendance.
- As an instructor, I want to end a session early if needed.

**Acceptance Criteria (Sprint 1 - UI Scope):**

- Dashboard renders the course list, session table, and flagged cases panel.
- Session table shows date, course, section, and time columns with sample data.
- Flagged cases panel shows student name, face match percentage, and Approve/Reject actions.
- 'Add New Session' button is visible and navigates to the session creation form.
- Dashboard is accessible only after role-based navigation from the login screen.

### 3.4 US-04 - Theme Switching (Light/Dark) - LEFTOVER

**Priority:** Medium

**Estimated Hours:** 5

**Status:** Leftover - Moved to Sprint 2

**User Story:**

As a user, I want to switch between light and dark themes so I can personalise how the app looks.

**Reason for Deferral:**

US-04 was deprioritized in Sprint 1 to focus development effort on completing the core UI screens (authentication, biometric enrollment, and all three dashboards). Theme switching will be implemented in Sprint 2 once the base UI is fully stable.

**Sub User Stories:**

- As a user, I want to select between light and dark themes.
- As a user, I want my theme choice to be remembered between sessions.
- As a user, I want the theme to apply consistently across the whole app.

**Acceptance Criteria (To be validated in Sprint 2):**

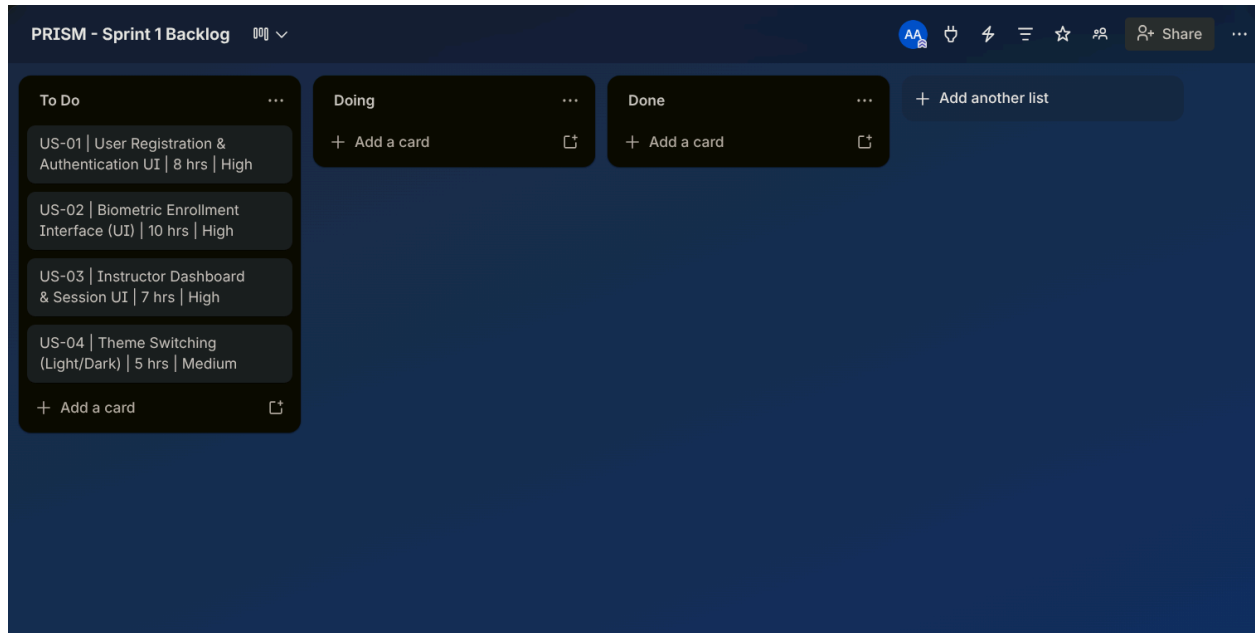
- Theme toggle is accessible from the Settings screen.
- Both light and dark options are available to all users.
- Selected theme persists across sessions.
- All UI components reflect the active theme without a full page reload.

## 4. Scrum Board (Trello)

The sprint backlog was uploaded to Trello and maintained throughout the sprint. Three snapshots were taken at different stages of the sprint.

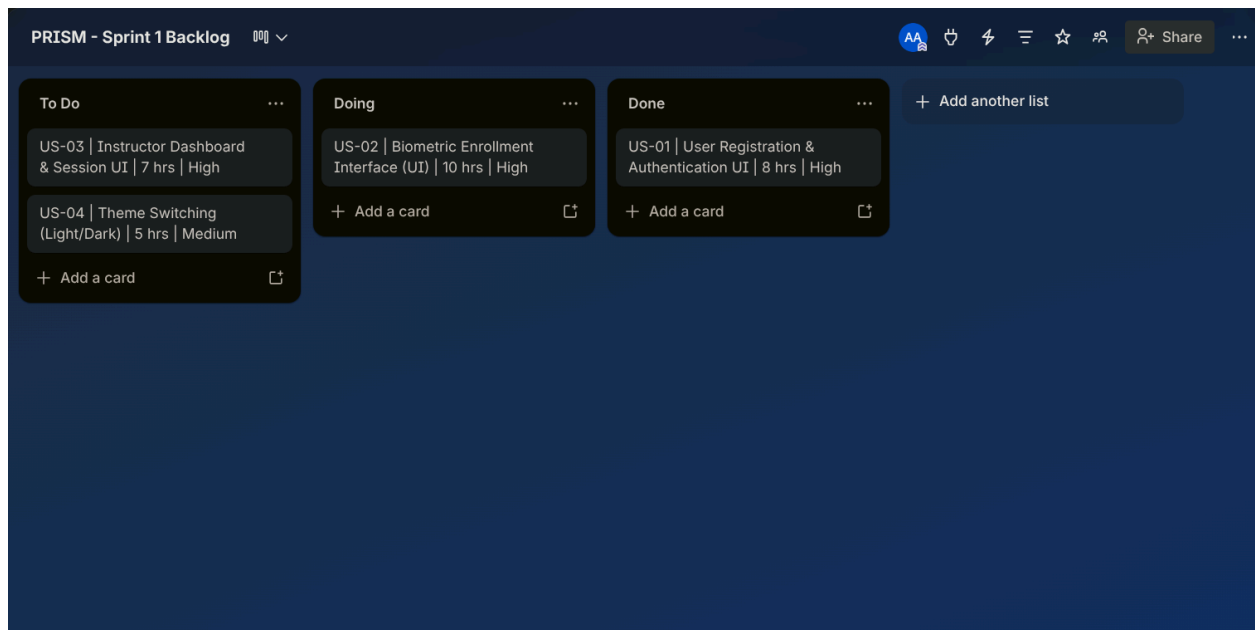
### Snapshot 1 - Sprint Start (Day 1)

All four user stories uploaded to the 'To Do' column. Sprint started with 30 estimated hours.



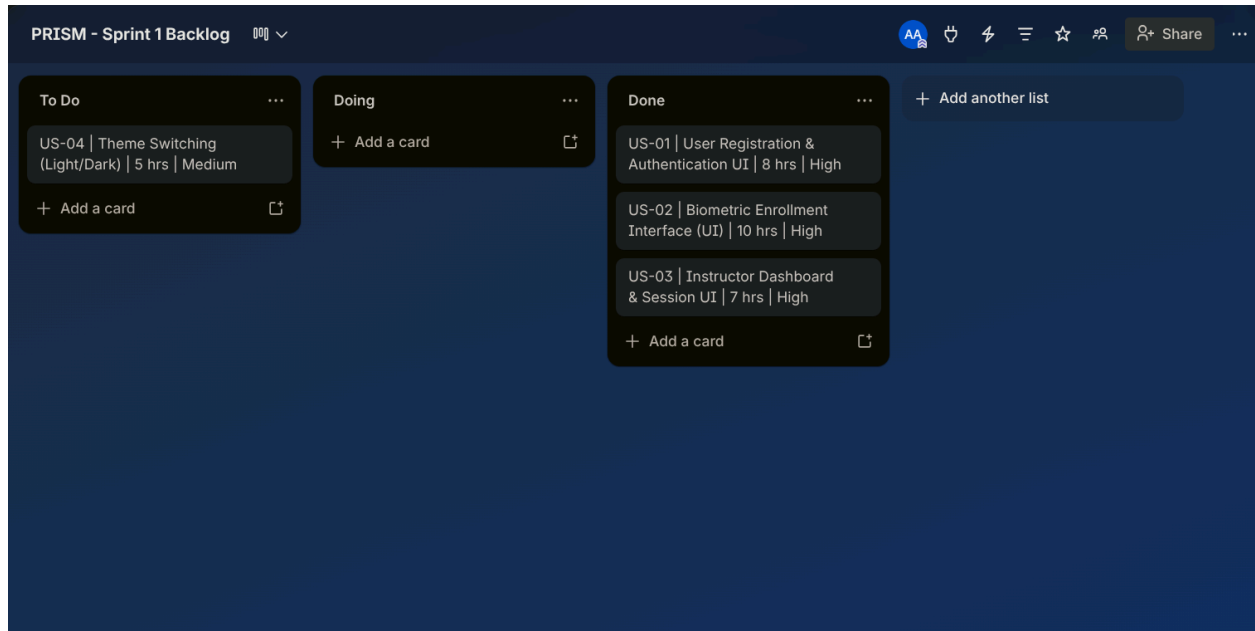
### Snapshot 2 - Mid-Sprint (Day 7)

US-01 completed and moved to 'Done'. US-02 moved to 'In Progress'. US-03 initiated. Around 50% of sprint work done.



### Snapshot 3 - Sprint End (Day 14)

US-01, US-02, and US-03 moved to 'Done'. US-04 (Theme Switching) marked as a Leftover and moved to Sprint 2.



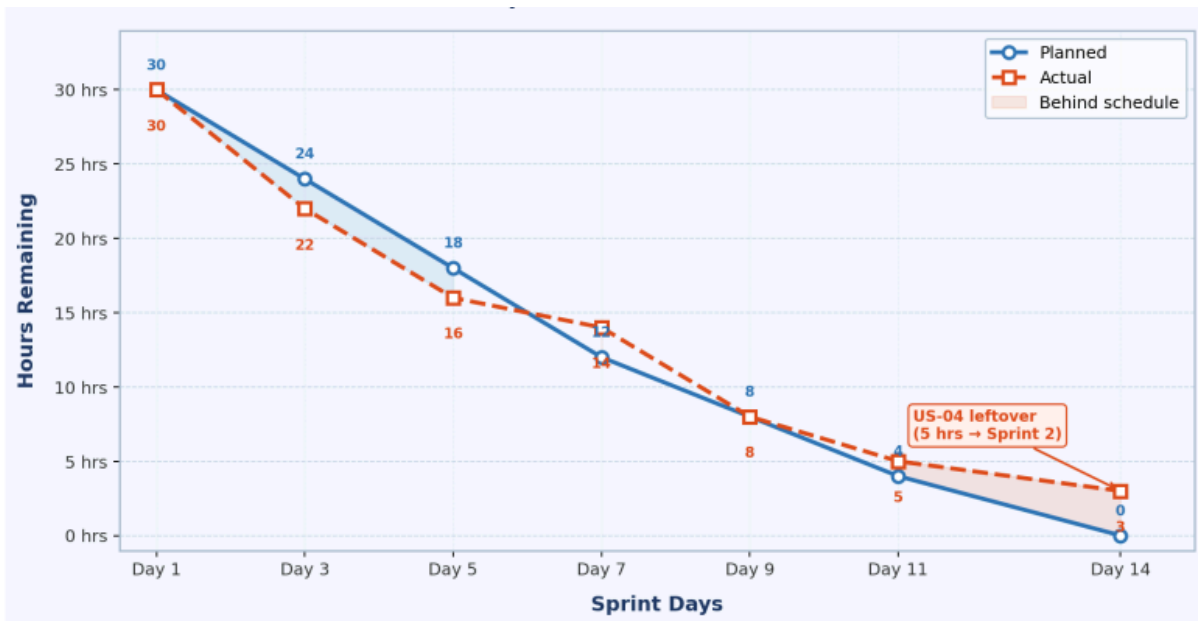


## 5. Product Burn-Down Chart - Sprint 1

The table below shows planned vs. actual hours remaining across the 14-day sprint.

| Day    | Planned Hours Remaining | Actual Hours Remaining | Notes                                 |
|--------|-------------------------|------------------------|---------------------------------------|
| Day 1  | 30                      | 30                     | Sprint start — all stories in backlog |
| Day 3  | 24                      | 22                     | US-01 auth UI in progress             |
| Day 5  | 18                      | 16                     | US-01 completed; US-02 started        |
| Day 7  | 12                      | 14                     | US-02 enrollment UI in progress       |
| Day 9  | 8                       | 8                      | US-03 dashboard UI in progress        |
| Day 11 | 4                       | 5                      | US-03 completed; US-04 started        |
| Day 14 | 0                       | 3                      | US-04 moved to Sprint 2 as leftover   |

*Note: US-04 (Theme Switching, 5 hours) was moved to Sprint 2 as the team focused on completing the core UI screens for authentication, biometric enrollment, and all role-based dashboards first.*



## 6. Non-Functional Requirements

The following non-functional requirements apply to the PRISM system:

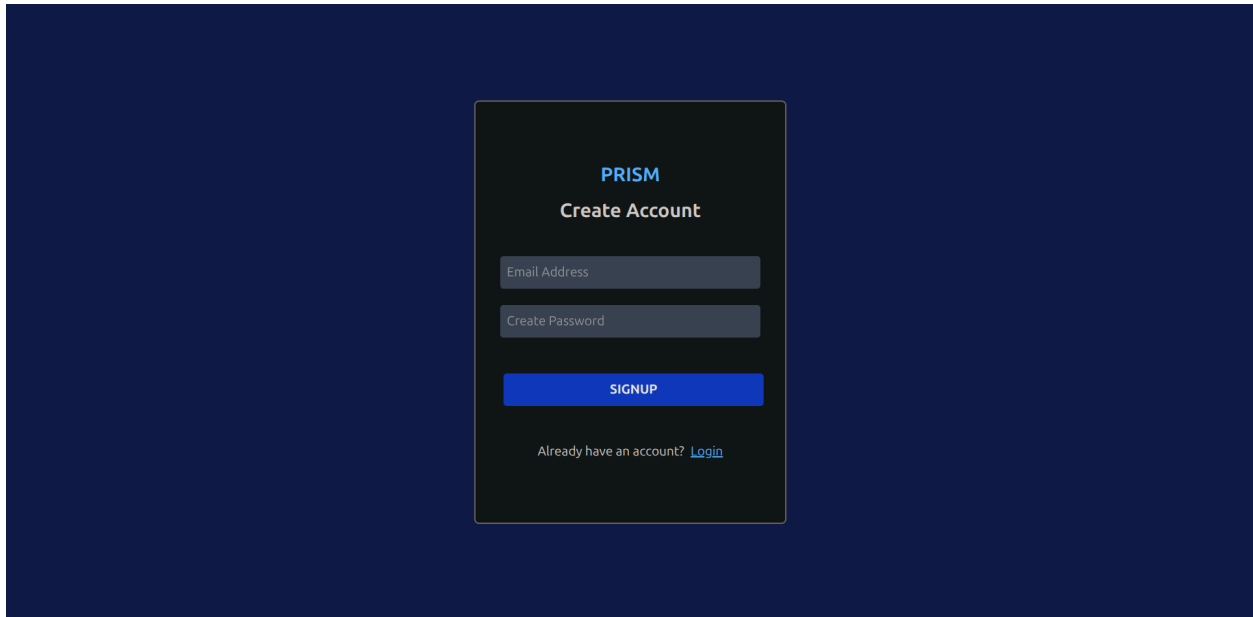
- **Security:** All user passwords will be stored using bcrypt hashing. Biometric data will be stored as encrypted templates only - no raw images will be saved. (To be implemented in Sprint 2.)
- **Performance:** Pages should load within 2 seconds on a standard connection. The React frontend has been structured with component-level code splitting to support this.
- **Usability:** The app works on both desktop and mobile browsers. All main actions are reachable within a few clicks from the dashboard. The UI has been validated across screen sizes in Sprint 1.
- **Reliability:** The system will aim for high uptime. If the backend goes down, users will see a clear error message rather than a blank screen. Error boundary components are planned for Sprint 2.
- **Inclusivity:** Students are never forced into a single biometric method. The enrollment UI explicitly offers face-only, fingerprint-only, or both, supporting students who may be unable to use facial recognition.

## 7. Implementation Screenshots

The following screenshots show the working frontend implementation at the end of Sprint 1. All screens are built in React with Tailwind CSS. Navigation between screens is fully functional. Data shown is hardcoded for UI validation; real data integration is planned for Sprint 2.

### Screenshot 1 - Sign Up Page

The user registration form with email, password, and role fields. Form validation UI is in place.

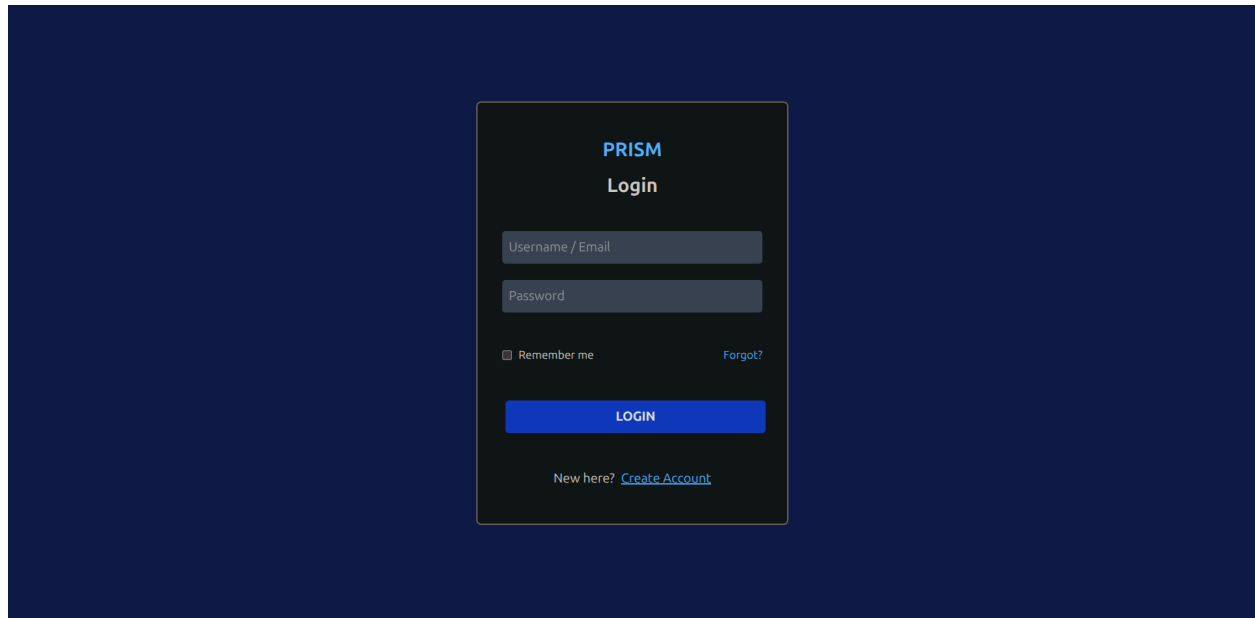


The screenshot displays a user registration form titled "PRISM Create Account" centered on a dark blue background. The form is contained within a dark gray box and includes the following elements:

- PRISM** (in blue) and **Create Account** (in white) at the top.
- An **Email Address** input field with a light gray border and placeholder text.
- A **Create Password** input field with a light gray border and placeholder text.
- A blue **SIGNUP** button.
- A link at the bottom: "Already have an account? [Login](#)".

## Screenshot 2 - Login Page

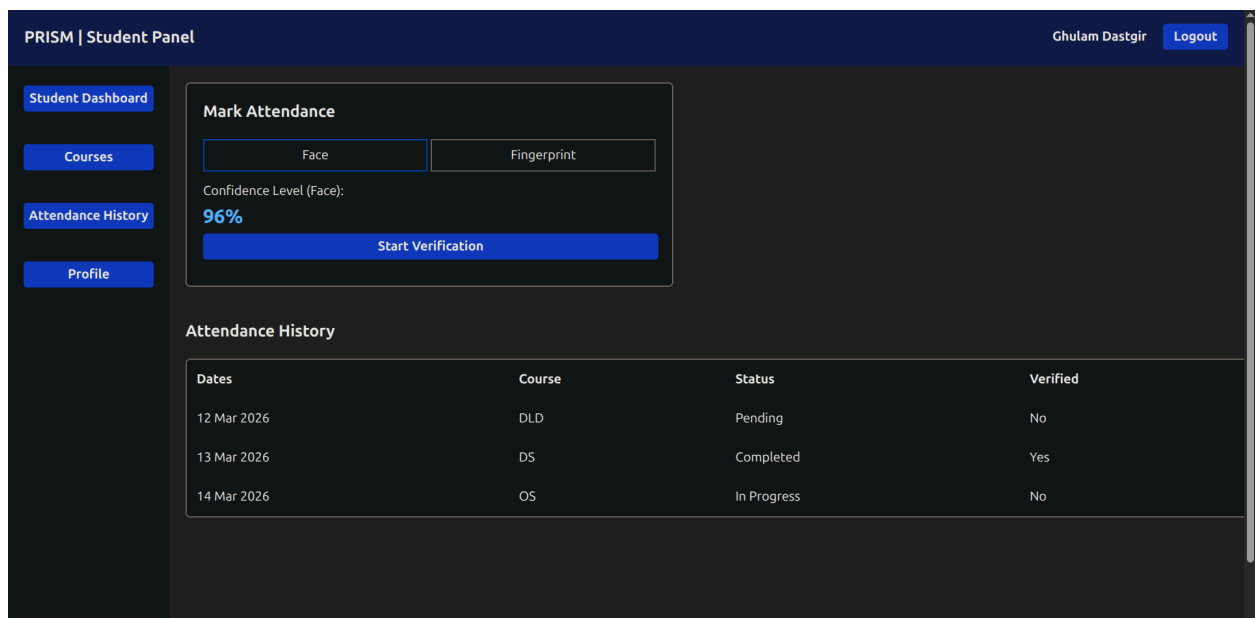
The login screen with username/email and password fields, a 'Remember me' checkbox, and a 'Forgot?' link.



The login page features a dark blue background with a central white login form. The form is titled 'PRISM Login' and contains fields for 'Username / Email' and 'Password'. Below these fields are a 'Remember me' checkbox and a 'Forgot?' link. A prominent blue 'LOGIN' button is positioned below the form. At the bottom of the form, there is a link for 'New here? Create Account'.

## Screenshot 3 - Student Dashboard

Student dashboard showing the attendance marking section with Face and Fingerprint tabs, a confidence level indicator (96%), a 'Start Verification' button, and an Attendance History table with course-wise records showing Present (Face), Present (Fingerprint), and Absent statuses.



The student dashboard is titled 'PRISM | Student Panel' and includes a user profile 'Ghulam Dastgir' with a 'Logout' button. The left sidebar contains navigation links for 'Student Dashboard', 'Courses', 'Attendance History', and 'Profile'. The main content area is divided into two sections: 'Mark Attendance' and 'Attendance History'.

**Mark Attendance Section:**

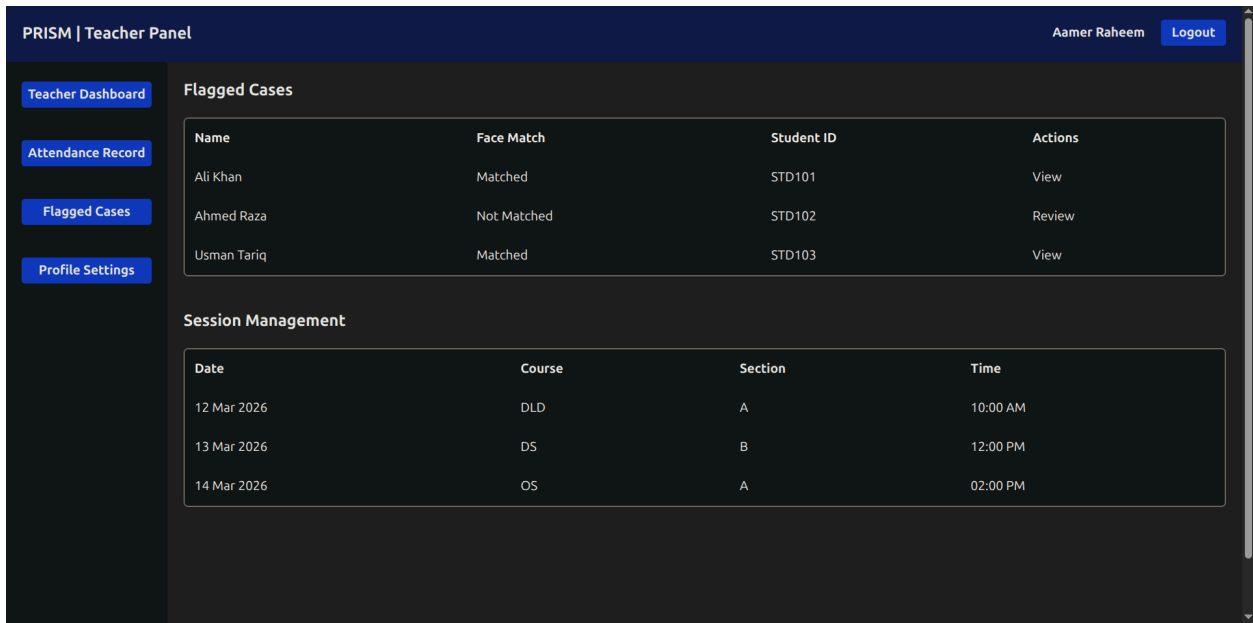
- Two tabs: 'Face' (selected) and 'Fingerprint'.
- Confidence Level (Face): **96%**
- A blue 'Start Verification' button.

**Attendance History Table:**

| Dates       | Course | Status      | Verified |
|-------------|--------|-------------|----------|
| 12 Mar 2026 | DLD    | Pending     | No       |
| 13 Mar 2026 | DS     | Completed   | Yes      |
| 14 Mar 2026 | OS     | In Progress | No       |

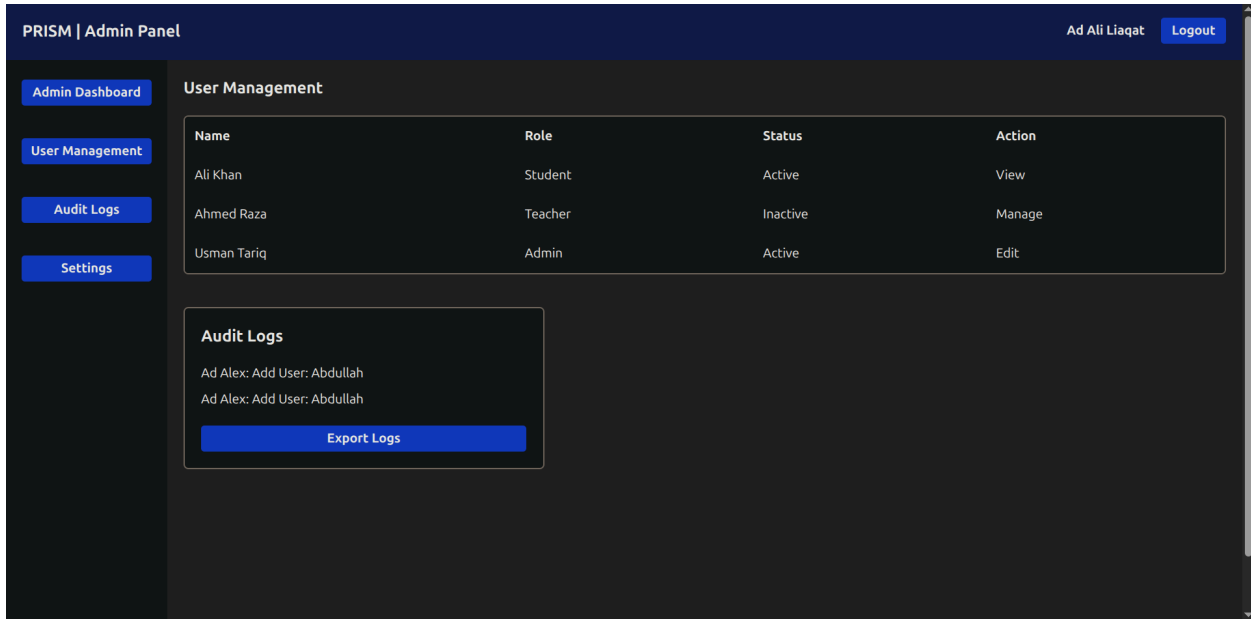
## Screenshot 4 - Teacher Dashboard

Instructor dashboard showing the Flagged Cases panel (students with low face match percentages and Approve/Reject actions) and the Session Management table with course, section, and time data. An 'Add New Session' button is present.



## Screenshot 5 - Admin Dashboard

Admin panel showing the User Management table (listing users by name, role, and status with Edit/Delete actions) and an Audit Logs panel showing recent system activity. An 'Export Logs' button is included.



## 8. Work Division

| Member      | Role                     | Responsibilities                                                                                                                                      |
|-------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Areeb Ahmed | Project Lead & Developer | System design, complete frontend development (React + Tailwind CSS), backend project scaffold (Node.js + Express + MongoDB connection), documentation |